SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA0614	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
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Section / Batch:	AI&DS	**Date:**	18-07-2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		**100 Marks**	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Video Compression System

A video compression system compares three algorithms with complexities O(n\^1.5),O(nlogn), and O(n\^2). Analyze all three using asymptotic notation. Compare their performance for high-resolution video data. Explain the impact of growth rates. Justify the most efficient algorithm.

Q2. IoT Data Processing Pipeline

An IoT system processes sensor data using the recurrence T(n)=3T(n/3)+nlog n. Apply the Master Theorem to determine the time complexity. Compare f(n)f(n)f(n) with nlog ban\^{\\log_b a}nlogba and identify the correct case. Analyze performance for large IoT datasets. Interpret efficiency.

Code of Conduct

I Mohan Kumar J [192424060] certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Mohan Kumar J*

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	**Understanding of Problem Context (20%)**	**Application of Concepts (30%)**	**Problem-Solving Approach (20%)**	**Accuracy of Solution (20%)**	**Clarity (10%)**	**Total Marks**
Q1 **(50 Marks)**						
Q2 **(50 Marks)**						
Overall Total (100)						

Faculty In-charge	**Course Coordinator**	**Head of Department**
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Question 1: Video Compression System

Problem Statement

A video compression system compares three algorithms with time complexities:

- Algorithm A: **O(n^1.5)**
- Algorithm B: **O(n log n)**
- Algorithm C: **O(n²)**

Analyze all three using asymptotic notation. Compare their performance for high-resolution video data. Explain the impact of growth rates and justify the most efficient algorithm.

Solution:

Step 1: Understanding Asymptotic Complexity

Asymptotic notation describes how an algorithm grows as the input size (**n**) increases.

Here,

- **Algorithm A:** O(n^1.5)
- **Algorithm B:** O(n log n)
- **Algorithm C:** O(n²)

A lower growth rate indicates a more efficient algorithm for large inputs.

Step 2: Compare Growth Rates

For very large values of **n**

$$n\log n<n^{1.5}<n^2$$

because

- log n grows very slowly.
- n^1.5 grows faster than n log n.
- n² grows the fastest.

Hence,

$$O(n\log n)<O(n^{1.5})<O(n^2)$$

Step 3: Numerical Comparison

Assume

$$n=1{,}000{,}000$$

Then,

Algorithm B

$$\mathbf{nlog_2\ 2}\boldsymbol{n}$$

$$=\mathbf{10^6 \times 20}$$

≈ 20 million operations

Algorithm A

$$\boldsymbol{n}^{\mathbf{1.5}}$$

$$=(\mathbf{10^6})^{\mathbf{1.5}}$$

$$=\mathbf{10^9}$$

= **1 billion operations**

Algorithm C

$$\boldsymbol{n}^{\mathbf{2}}$$

$$=(\mathbf{10^6})^{\mathbf{2}}$$

$$=\mathbf{10^{12}}$$

= **1 trillion operations**

Step 4: Comparison Table

Algorithm	**Time Complexity**	**Growth Rate**	**Performance for High-Resolution Video**
A	O(n^1.5)	Medium	Better than $O(n^2)$ but slower than O(n log n)
B	O(n log n)	Slowest	Best performance and scalability
C	$O(n^2)$	Fastest growth	Poor performance for large videos

Step 5: Impact of Growth Rate

High-resolution videos contain millions of pixels and frames.

As input size increases,

- O(n log n) increases slowly.
- O(n^1.5) increases much faster.
- $O(n^2)$ increases extremely rapidly.

Therefore,

- Processing time,
- CPU utilization,
- Memory access,
- Energy consumption

are all lowest for **O(n log n)**.

Step 6: Justification

The **O(n log n)** algorithm is the most efficient because:

- It has the smallest asymptotic growth.
- It scales well for very large videos (4K/8K).
- It requires significantly fewer operations.
- It provides faster compression with better resource utilization.

Final Answer

The order of efficiency is

$$\boxed{O(nlogn)<O(n1.5)<O(n2)}$$

Hence, the **O(n log n)** algorithm is the best choice for high-resolution video compression because it offers the lowest growth rate, executes faster for large datasets, and scales efficiently.

Q2 : IoT Data Processing Pipeline

Problem Statement

The IoT system follows the recurrence relation

T(n)=3T($\frac{n}{3}$)+nlogn

Apply the **Master Theorem** to determine the time complexity.

Compare

- f(n)
- $\log_b a$
- $n^{\log_b a}$

Identify the correct Master Theorem case, analyze performance for large IoT datasets, and interpret efficiency.

Solution

Step 1: Identify Parameters

The recurrence is

$$T(n)=aT(\frac{n}{b})+f(n)$$

where

- a=3
- b=3
- f(n)=nlog n

Step 2: Compute

$$n^{\log_b a}$$

Substitute:

$$a=3, b=3$$

$$\log_3 3=1$$

Therefore,

$$n^{\log_3 3}=n^1=n$$

So,

$$n^{\log_b a}=n$$

Step 3: Compare f(n) and $n^{\log_b a}$

We have

$$f(n)=n \log n$$

$$n^{\log_b a}=n$$

Clearly,

$$n \log n=\Theta(n log^1 n)$$

Step 4: Apply Master Theorem

Since

$$f(n)=\Theta(n^{\log_b a} log^k n)$$

where

- (k=1)

This satisfies **Master Theorem Case 2**.

According to Case 2,

$$T(n)=\Theta(n^{\log_b a} log^{k+1})$$

Substitute the values,

$$T(n)=\Theta(n log^2 n)$$

Step 5: Final Time Complexity

$$\mathbf{T(n)=\Theta(n log^2 n)}$$

Step 6: Performance Analysis

For large IoT datasets,

- Data is divided into **3 equal parts**.
- Each subproblem is solved recursively.
- Results are combined in **O(n log n)** time.

The overall complexity

$$\mathbf{\Theta(n log^2 n)}$$

is only slightly larger than linear time, making it efficient for processing millions of IoT sensor readings.

Step 7: Efficiency Interpretation

Advantages:

- Efficient divide-and-conquer strategy.
- Good scalability for increasing sensor data.
- Suitable for real-time IoT analytics.
- Lower computational cost than quadratic algorithms.
- Handles large-scale distributed processing effectively.

Recursion Tree Verification (Optional Justification)

At every level,

$$3^i$$

subproblems exist.

Each subproblem size is

$$\frac{\boldsymbol{n}}{\boldsymbol{3^i}}$$

Cost at level i:

$$3^i(\frac{\boldsymbol{n}}{\boldsymbol{3^i}}\mathbf{log}\frac{\boldsymbol{n}}{\boldsymbol{3^i}})=\text{n(log n-i)}$$

Summing across all $\log_3 n$levels gives

$$\Theta(\text{n}log^2\text{n}),$$

which confirms the Master Theorem result.

Final Answer

Given

$$\text{T(n)=3T}(\frac{n}{3})\text{+nlogn},$$

- a=3, b=3, f(n)=nlog n
- $n^{\log_b a}$=n
- f(n)=Θ(nlogn)=$\Theta(n^{\log_b a} log^1 n)$

Therefore, **Master Theorem Case 2** applies.

$$\text{T(n)}=\Theta(\text{n}log^2\text{n})$$

This complexity is efficient for large IoT data processing because it scales well with increasing data volume while maintaining significantly better performance than quadratic-time algorithms.